STANFORD UNIVERSITY GRADUATE SCHOOL OF BUSINESS

MASTER IN BUSINESS ADMINISTRATION

OIT 367.1 – BUSINESS INTELLIGENCE FROM BIG DATA

Final Project:
Predicting the success of startups at the Series A stage

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(Yes, you may use our report)

1. EXECUTIVE SUMMARY

1.1 Introduction

In this report, we work with data provided by Thomvest Ventures ("Thomvest") to explore potential predictors of success for startup companies. Thomvest is a venture capital firm that invests in early stage startups (pre-seed / seed / series A) primarily in the fintech, real estate technology, cybersecurity / cloud infrastructure, and sales & marketing technology sectors.[1]

Thomvest is particularly interested in what quantitative and qualitative factors predict startup success, and which investors at the seed stage are most likely to fund successful companies. If we are able to identify a collection of predictors based on available datapoints (e.g., number of prior funding rounds, total funding to date, "quality" of prior investors, etc.), it would be valuable to not only Thomvest but all venture capital / growth-stage investment firms. A predictive model would help investors make smarter, more informed investments decisions and lead to higher returns. Additionally, a quantitative model could help automate investment sourcing and diligence, processes that are currently time-consuming and headcount-heavy.

1.2 Problem Statement

Ultimately, we would like to answer the following question for Thomvest:

Are there certain qualitative and/or quantitative metrics that can accurately predict the success of startups, where success is defined as raising an additional round of capital from a high-quality investor? Which factors are most predictive of future investment?

1.3 Principal Findings and Recommendations

Based on our analysis, the following metrics are most predictive of a startup's ability to raise a future round: number of days between the Series A and prior round; number of investors in prior rounds; total funding up to the Series A; if the company has received an investment from a top tier investor in a prior round; if the company is based in the Bay Area; and if the company is in the software, IT and science & engineering sector. These insights are valuable in helping to guide future investment decisions.

Importantly, we recommend looking at a fixed investment window when determining "success" to avoid comparing different time horizons among different startup cohorts.

[1] Sector focuses based on Thomvest's LinkedIn profile.

Additionally, we did not want to over penalize companies that did not raise an additional round due to lack of need for funding.

For future studies, we believe the results could be greatly improved if the detailed data on social media follower and like counts, number of employees, etc. were provided at the time of each funding round. We would also recommend trying to compile additional data on post-money valuation of prior rounds.

2. DATA

2.1 Data Overview

We received raw data directly from Thomvest in the form of 4 key spreadsheets.

Company details. The primary file includes information on 1,033 companies, all of which meet the following criteria: U.S.-based companies that raised a Series A or B in the last two years for which Thomvest has near-complete data on financing history and investors. For each company, Thomvest provided the following datapoints:

- Website and social media URLs
- Short company description
- Industry / sector
- Location
- Founding date
- Number of founding rounds
- Total funding
- Last funding round date
- Approximate employee count
- Contact information

Funding round details. For each of the 1,033 selected companies, Thomvest also provided publicly available data on each funding round to date (sourced from Crunchbase). There were a total of 5,578 rows / funding rounds included, each including the following:

- Company name
- Investment stage
- Number of investors
- Funding announcement date
- Funding amount (for 80% of the data)
- List of investors
- Lead investor
- Crunchbase source link

Investor tiering. Thomvest categorizes the list of other investors into 3 tiers, with 1 being the "highest quality" and 3 being the "lowest quality." This internal ranking is based on Thomvest's prior working history with each firm and each firm's track record for making strong investments. Of the 382 investors included in Thomvest's investor file / represented in the 5,578 funding rounds, 57 (~15%) are ranked as Tier 1, 275 (~72%) as Tier 2, and 50 (~13%) as Tier 3.

Qualitative metrics. Finally, Thomvest also shared the following metrics in a JSON file:

- Traction metrics including Twitter, Facebook and Instagram follower / like counts
- Customer type: e.g., B2B vs. B2C
- Qualitative highlights generated by Thomvest's data partner Harmonic.ai (e.g., "seasoned executive," "top university," "seasoned operator")

2.2 Defining "Success"

With regards to data pre-processing, the most important step was to define "success." We treated our problem statement as a standard classification problem, with "success" as the outcome / response variable. To start, we took into account Thomvest's definition of success: *"raising a new round of venture capital, ideally from a top-tier venture firm."*

Startups raise a new round of funding every 12-24 months, on average.[2] Thus, to avoid unfairly classifying a company that has not raised an additional round due to timing / lack of capital need as a "failure," we chose to look at a fixed funding window. We set the window "starting time" as the date of the Series A investment, as Thomvest provided complete Series A data for all companies in our dataset.

Accordingly, we classified all companies that had successfully raised a Series B round from a Tier 1 investor within the funding window as a "success." For consistency, we removed all companies for which the time between the Series A investment and today (March 2022) was less than the length of our funding window from our analysis. In summary, we sorted companies into the following 3 categories:

- **"Success" / outcome value of 1**
  - Companies that raised a Series B round from a *Tier 1 investor within the funding window*
- **"Failure" / outcome value of 0**
  - Companies that raised a Series B round from *only Tier 2 and 3 investors* within the funding window
  - Companies that have *not yet raised a Series B within the funding window*
- **Disregarded / excluded from analysis**
  - Companies for which their Series A investment was too recent

[2] https://visible.vc/blog/startup-funding-stages/

We explored various funding windows in our analysis. Below, we summarize the (1) percentage of companies eligible and (2) percentage of companies classified as a "success" for each window. As we increase the length of the window, the number of eligible companies (sample size) decreases, and the percentage of successful companies increases as we have given them more time to raise from a top-tier investor.

	# Months						
	12	18	24	30	36	48	60
% of companies Eligible	80%	71%	57%	52%	45%	30%	19%
Out of eligibles, % that is successful	3%	6%	12%	16%	21%	40%	67%

Total Companies = 1,033

Definition of companies Eligible: Companies that have raised a Series A in less than X months ago from today (# of months in the column)

% that is successful: Companies that have raised Series B from a Top Tier investor in the # of months in the column after the Series A

2.3 Additional Pre-Processing / Feature Engineering

First, we determined what factors we wanted to analyze. We ultimately decided on the following variables – a combination of the provided raw variables and new features.

Variable Name	**Type**	**Description**
Dayspassed	Numerical	Number of days between company founding and Series A round date
Date between A and previous	Numerical	Number of months between Series A & most recent prior funding round
# Investors round before Series A	Numerical	Number of investors in round prior to the Series A
# Investors on Series A	Numerical	Number of investors in Series A round
Funding up to Series A	Numerical	Total funding in rounds prior to Series A
Top Tier Investor in Previous Round	Binary	If a Tier 1 investor has invested in any round prior to the Series B
Sector	Categorical	Company's industry sector(s) - each company can have multiple values
City	Categorical	City / state of headquarters
Success	Binary	If a company has raised a Series B round from a Tier 1 investor

Note we chose to exclude a subset of the provided data including traction metrics (social media follower and like counts) and employee headcount due to inconsistent timing and success bias. These metrics were only provided as of "today" – not at the

time of each company's Series A. Thus, including these current variables in our regression would inflate the predictive power of our model (given that "successful" companies are likely to have higher traction metrics by virtue of still being around, and even more so for those that raised their round many years ago).

Numerical variables: Using the provided data, we calculated several numerical variables that we felt were more predictive than the raw data. For example, we added a variable "Years," which is the number of years between the founding date of the company and its Series A announcement. These variables can roughly be categorized into timing (years, date between A and previous), funding size (funding up to Series A), and investors (number of investors round before Series A, number investors on Series A, top tier investor in previous round). We consolidated information from the funding and companies files in Excel, and manipulated the data to create our new variables.

Categorical variables: We converted our 2 categorical variables – sector and location – into numerical variables through binning. First, however, we needed to clean the sector information, as the provided excel included multiple sector names in the same cell (e.g., Company A's category column included 3 sectors, separated by commas: "Software, Travel, Apps"). First, we separated the multi-sector names into different columns. Then, we created dummy variables for each column that had similar categories. The categorical variable "city" was easier to incorporate – we converted these directly to dummy variables. We also created a new binary variable for whether a Tier 1 investor had invested in a round prior to the Series B. In Excel, we also used the funding dataset to determine what top tier funds had invested in these companies in prior rounds so we could evaluate which individual fund (e.g., Andreessen Horowitz, Sequoia, etc.) is best at "picking" successful companies.
We combined all of these variables into 1 spreadsheet using unique identifiers (i.e., company name and investor name) and appended our new "success" variable to the final dataset as a new column.

2.4 Data Limitations

It is also important to note several limitations with the provided data. One of the largest challenges we faced was "sampling on the dependent variable." As the dataset only includes companies that successfully raised a Series A (for which Thomvest had near-complete data), we lose insights from true "failures" – companies that raised an angel and/or seed round and failed to raise a Series A / are no longer in operation. For that reason, we selected the Series A as our starting point given that all companies in the dataset had raised a Series A (based on selection criteria).

Another limitation was that some of the provided data were measured as of “today” (e.g., followers / likes on social media). As our analysis assumes an investment decision at the time of the Series A, these metrics would technically be “future” data points not available at the time of the analysis. Additionally, such data is impacted by the company’s success.

In terms of potential predictor variables, we would ideally have access to other quantitative metrics including post-money valuation of each round / corresponding step-up values, revenue growth estimates, market share metrics, etc. Moreover, the most accurate metric of “success” of an investment from the perspective of venture capital firms such as Thomvest is that company’s investment return (IRR / MOIC). Unfortunately, financial, valuation and returns data are highly sensitive for early stage companies and generally not publicly disclosed.

3. METHODOLOGY

For our analyses, we fitted the following three types of models for each funding window length (ranging from 1 to 5 years): tree, random forest and GBM. We then compared the resulting AUCs to evaluate the predictive power of each model. To strengthen our model, we also performed ten-fold cross validation and calculated the average AUC across all folds. For the selected model, we used the “importances” function in python to better understand the selected decision variables.

4. ANALYSIS / RESULTS

Of the three models we trained, the Random Forest performed the best on our validation set, with an AUC of 0.70 for both the 3-year and 5-year funding windows. We used 3 years as our baseline because it had a larger sample set and still guaranteed we were close to the average 12-24 month funding window.

	Model AUC		
Length of Funding Window	Tree	Random Forest	GBM
12 Months (1 Year)	0.50	0.45	0.41
18 Months (1.5 Years)	0.49	0.62	0.64
24 Months (2 Years)	0.50	0.67	0.61
30 Months (2.5 Years)	0.60	0.64	0.65
36 Months (3 Years)	**0.62**	**0.70**	**0.67**
48 Months (4 Years)	0.58	0.60	0.60
60 Months (5 Years)	0.61	0.70	0.69

We were also able to gain some additional qualitative insights from our GBM model. Below, we summarized the top feature rankings. The most predictive variables were: number of days between founding and Series A, number of investors prior to the Series A, date between the Series A and prior round of funding.

```
Feature rankings:
1. Dayspassed (0.108613)
2. # Investors round before Series A (0.090967)
3. Date between A and previous (0.081498)
4. # Investors on Series A (0.075780)
5. Funding Up to Series A (0.070516)
6. Top Tier Investor in Previous Round (0.029127)
7. Palo Alto (0.027395)
8. Software (0.025482)
9. Redwood City (0.023387)
10. San Francisco (0.023086)
11. Andreessen Horowitz (0.020056)
12. Information Technology (0.018214)
13. New York (0.018213)
14. Lakewood (0.017813)
15. Science and Engineering (0.016747)
```

5. CONCLUSION

There are many factors that contribute to the success of companies, such as the founding team, macroeconomic environment, market trends and competitive dynamics (e.g., SVB's failure will likely affect valuations) - thus, it was challenging to create an extremely predictive model based on the available data. However, we believe that we found meaningful insights that could be valuable alongside the firm's usual diligence process.

As described above, we picked a fixed funding window to define success so as not to penalize companies that use capital more efficiently and take more time to raise subsequent rounds. Moreover, we did not use all data points provided (e.g. social media likes) due to timing biases. These metrics would be interesting to analyze in future studies.

In conclusion, our model found that the key decision variables were the time between the founding date and Series A, number of investors, time between the Series A and prior round, having a top tier investor in a prior round, and being located in the Bay Area.

6. APPENDIX

Note: see attached files for python code.

DATA CLEANING AND CREATION OF NEW VARIABLES

The images below show the tables we compiled in excel to create some of our new variables based on provided inputs (light blue), organized by funding round.

Seed	Series A	Series B	org_name	Month announced	Months since 2023-03	name	investors	Sum of raised_amount_usd	Series A?	Series B?
0x0	0x0	0x0	0x	2017-01	-74	Venture Round - 0x	Boost VC		0	0
0x0	0x0	0x0	0x	2017-06	-69	Initial Coin Offering - 0x	Andreas Schwartz, Ca	24,000,000	0	0
0x0	0x0	0x0	0x	2017-07	-68	Seed Round - 0x	8 Decimal Capital, AlphaCoin Fund, Fabric \		0	0
0x0	0x0	0x0	0x	2017-07	-68	Seed Round - 0x	Hex Capital, Kenetic		0	0
0x0	0x0	0x0	0x	2017-10	-65	Venture Round - 0x	Placeholder		0	0
0x0	0x0	0x0	0x	2020-10	-29	Secondary Market - 0x	(blank)		0	0
0x1	0x0	0x0	0x	2020-10	-29	Venture Round - 0x	Bossa Nova Investimentos		0	0
0x0	0x1	0x0	0x	2021-02	-25	Series A - 0x	Blockchain.com Vent	14,999,960	1	0
0x0	0x0	0x1	0x	2022-04	-11	Series B - 0x	A.Capital Ventures, A	70,000,000	0	1
2UL	2UL	2UL	2ULaundry	2016-11	-76	Pre Seed Round - 2ULau	(blank)	400,000	0	0
2UL	2UL	2UL	2ULaundry	2017-07	-68	Pre Seed Round - 2ULau	Techstars	20,000	0	0
2UL	2UL	2UL	2ULaundry	2017-09	-66	Seed Round - 2ULaundr	Techstars	2,500,000	0	0
2UL	2UL	2UL	2ULaundry	2018-10	-53	Seed Round - 2ULaundr	Techstars	300,000	0	0
2UL	2UL	2UL	2ULaundry	2019-09	-42	Series A - 2ULaundry	Engage Ventures, Ref	6,000,000	1	0
2UL	2UL	2UL	2ULaundry	2023-01	-2	Series B - 2ULaundry	Level 5 Capital Partn	20,000,000	0	1
3AM	3AM	3AM	3AM Innovations	2017-01	-74	Pre Seed Round - 3AM I	Launch NY, Trend Dis	1,052,500	0	0
3AM	3AM	3AM	3AM Innovations	2018-04	-59	Convertible Note - 3AM	Launch NY		0	0
3AM	3AM	3AM	3AM Innovations	2019-03	-48	Convertible Note - 3AM	Launch NY	1,150,000	0	0
3AM	3AM	3AM	3AM Innovations	2019-05	-46	Convertible Note - 3AM	Launch NY		0	0
3AM	3AM	3AM	3AM Innovations	2019-07	-44	Convertible Note - 3AM	Urban Us		0	0
3AM	3AM	3AM	3AM Innovations	2019-09	-42	Pre Seed Round - 3AM I	Urban Us, URBAN-X		0	0
3AM	3AM	3AM	3AM Innovations	2020-01	-38	Grant - 3AM Innovation	BIRD Foundation		0	0
3AM	3AM	3AM	3AM Innovations	2020-04	-35	Venture Round - 3AM Ir	TechNexus Venture Collaborative		0	0
3AM	3AM	3AM	3AM Innovations	2020-06	-33	Series A - 3AM Innovatic	Launch NY	25,000	1	0
3AM	3AM	3AM	3AM Innovations	2020-08	-31	Seed Round - 3AM Inno	TechNexus Venture Collaborative		0	0
3AM	3AM	3AM	3AM Innovations	2021-04	-23	Venture Round - 3AM Ir	TechNexus Venture Collaborative		0	0
3AM	3AM	3AM	3AM Innovations	2022-08	-7	Series A - 3AM Innovatic	Impellent Ventures, (	9,000,000	1	0
3AM	3AM	3AM	3AM Innovations	2022-11	-4	Secondary Market - 3AM	Outbound Capital		0	0
3DL	3DL	3DL	3DLOOK	2016-05	-82	Angel Round - 3DLOOK	Vadim Rogovskiy	100,000	0	0
3DL	3DL	3DL	3DLOOK	2017-08	-67	Convertible Note - 3DLO	Alex Bornyakov, Boo	305,000	0	0
3DL	3DL	3DL	3DLOOK	2018-07	-56	Seed Round - 3DLOOK	500 Startups, u.ventu	1,000,000	0	0
3DL	3DL	3DL	3DLOOK	2019-10	-41	Seed Round - 3DLOOK	Anton Borzov, Micha	1,750,000	0	0
3DL	3DL	3DL	3DLOOK	2020-03	-36	Convertible Note - 3DLO	NETWORK VC		0	0
3DL	3DL	3DL	3DLOOK	2020-07	-32	Seed Round - 3DLOOK	Solid5 Ventures, ICU	1,500,000	0	0
3DL	3DL	3DL	3DLOOK	2021-11	-16	Series A - 3DLOOK	Flyer One Ventures, F	10,000,000	1	0
3T B	3T B	3T B	3T Biosciences	2017-02	-73	Convertible Note - 3T Bi	Asset Management Ventures (AMV)		0	0

Series A more than 2 yrs ago?	Series B Top tier?	Round prior to Series A	# Investors	Funding Accumulated	Top Tier investor acc.	Top Tier investment	Goldman Sachs Principal Strategic Investments	8VC	Sapphire Ventures	Benchmark	Lightspeed Venture Partners	Khosla Ventures	Greylock Partners
		0	1	0	0	0	0	0	0	0	0	0	0
		0	11	24,000,000	0	0	0	0	0	0	0	0	0
		0	5	24,000,000	0	0	0	0	0	0	0	0	0
		0	2	24,000,000	0	0	0	0	0	0	0	0	0
		0	1	24,000,000	0	0	0	0	0	0	0	0	0
		0	1	24,000,000	0	0	0	0	0	0	0	0	0
		1	1	24,000,000	0	0	0	0	0	0	0	0	0
0		0	7	38,999,960	0	0	0	0	0	0	0	0	0
	1	0	14	108,999,960	1	1	0	0	0	0	0	0	1
		0	1	400,000	0	0	0	0	0	0	0	0	0
		0	1	420,000	0	0	0	0	0	0	0	0	0
		0	1	2,920,000	0	0	0	0	0	0	0	0	0
		1	1	3,220,000	0	0	0	0	0	0	0	0	0
0		0	6	9,220,000	0	0	0	0	0	0	0	0	0
	0	0	1	29,220,000	0	0	0	0	0	0	0	0	0
		0	3	1,052,500	0	0	0	0	0	0	0	0	0
		0	1	1,052,500	0	0	0	0	0	0	0	0	0
		0	1	2,202,500	0	0	0	0	0	0	0	0	0
		0	1	2,202,500	0	0	0	0	0	0	0	0	0
		0	1	2,202,500	0	0	0	0	0	0	0	0	0
		0	2	2,202,500	0	0	0	0	0	0	0	0	0
		0	1	2,202,500	0	0	0	0	0	0	0	0	0
		1	1	2,202,500	0	0	0	0	0	0	0	0	0
0		0	1	2,227,500	0	0	0	0	0	0	0	0	0
		0	1	2,227,500	0	0	0	0	0	0	0	0	0
		1	1	2,227,500	0	0	0	0	0	0	0	0	0
0		0	2	11,227,500	0	0	0	0	0	0	0	0	0
		0	1	11,227,500	0	0	0	0	0	0	0	0	0
		0	1	100,000	0	0	0	0	0	0	0	0	0
		0	3	405,000	0	0	0	0	0	0	0	0	0
		0	2	1,405,000	0	0	0	0	0	0	0	0	0
		0	4	3,155,000	0	0	0	0	0	0	0	0	0
		0	1	3,155,000	0	0	0	0	0	0	0	0	0
		1	2	4,655,000	0	0	0	0	0	0	0	0	0
0		0	5	14,655,000	0	0	0	0	0	0	0	0	0
		0	1	0	0	0	0	0	0	0	0	0	0

The two images below show a compilation of the above data, organized by company.

Company	Company_adjusted	Series A?	Series A more than x yrs ago?	Series B?	Series B with top tier?	Series B in less than 24 months?	Type of investment	Success	Date Series A
0x	0x	1	0	1	1	1	A. Not enough time	-1	2021-02
2ULaundry	2ULaundry	1	0	1	0	0	A. Not enough time	-1	2019-09
3AM Innovations	3AM Innovations	1	0	0	0	0	A. Not enough time	-1	2020-06
3DLOOK	3DLOOK	1	0	0	0	0	A. Not enough time	-1	2021-11
3T Biosciences	3T Biosciences	1	0	0	0	0	A. Not enough time	-1	2022-08
410 Medical	410 Medical	1	1	1	0	0	C. Raised non-top-tier Series B	0	2017-03
75F	75F	1	0	0	0	0	A. Not enough time	-1	2019-09
Abound	Abound	1	0	1	0	0	A. Not enough time	-1	2021-02
AccelByte	AccelByte Inc	1	0	1	0	0	A. Not enough time	-1	2021-08
Accern	Accern	1	0	1	0	0	A. Not enough time	-1	2020-03
Access Vascular	Access Vascular	0	0	1	0	0	A. Not enough time	-1	
Aclaimant	Aclaimant	1	0	1	0	0	A. Not enough time	-1	2020-06
Activ Surgical	Activ Surgical	1	0	1	0	0	A. Not enough time	-1	2019-10
Adaptive Phage The	Adaptive Phage The	0	0	1	0	0	A. Not enough time	-1	
Adlumin	Adlumin, Inc	0	0	0	0	0	A. Not enough time	-1	
Adlumin, Inc	Adlumin, Inc	1	0	1	0	0	A. Not enough time	-1	2020-09
Adranos Energetics	Adranos Energetics	1	0	0	0	0	A. Not enough time	-1	2022-04
Advocado	Advocado	1	0	0	0	0	A. Not enough time	-1	2021-12
AEENA Dx	AEENA Dx	1	0	0	0	0	A. Not enough time	-1	2020-05
Aerami Therapeutic	Aerami Therapeutic	0	0	1	0	0	A. Not enough time	-1	
Aerones	Aerones	1	0	0	0	0	A. Not enough time	-1	2022-12
Agent IQ	Agent IQ	1	1	0	0	0	B. Didn't raise Series B	0	2018-02
Agility Robotics	Agility Robotics	1	0	1	0	0	A. Not enough time	-1	2018-03
AgroSpheres	AgroSpheres	1	0	1	0	0	A. Not enough time	-1	2019-08
AgVend	AgVend	1	0	0	0	0	A. Not enough time	-1	2022-03
AiFi	AiFi	1	0	1	0	0	A. Not enough time	-1	2019-01
Aikon	Aikon	1	0	0	0	0	A. Not enough time	-1	2022-09
Air	Air	1	0	0	0	0	A. Not enough time	-1	2020-09
Akadeum Life Scien	Akadeum Life Scien	1	0	1	0	0	A. Not enough time	-1	2019-12
Alcatraz AI	Alcatraz AI	1	0	0	0	0	A. Not enough time	-1	2022-09
Aliaswire	Aliaswire	1	0	0	0	0	A. Not enough time	-1	2021-06
ALICE Technologies	Alice Technologies	1	0	1	0	0	A. Not enough time	-1	2019-05
Alitheon	Alitheon	1	0	0	0	0	A. Not enough time	-1	2022-09
Alkymi	Alkymi	1	0	0	0	0	A. Not enough time	-1	2022-10
Allganize	Allganize	1	0	0	0	0	A. Not enough time	-1	2019-04

Date Series A	Date Round prior to Series A	Date between A and previous	# Investors round before Series A	# Investors on Series A	Funding Up to Series A	Top Tier Investor in Previous Round	Date Series B	Time between Series A & B	Goldman Sachs Principal Strategic Investments	Blackstone
2021-02	2020-10	4	1	7	24,000,000	0	2022-04	14	0	0
2019-09	2018-10	11	1	6	3,220,000	0	2023-01	40	0	0
2020-06	2020-04	2	1	1	2,202,500	0			0	0
2021-11	2020-07	16	2	5	4,655,000	0			0	0
2022-08	2017-10	58	7	2	12,000,000	1			0	0
2017-03	2016-09	6	1	3	2,200,000	0	2022-11	68	0	0
2019-09	2018-06	15	2	6	2,105,000	0			0	0
2021-02	2018-02	36	1	3	4,180,000	0	2021-11	9	0	0
2021-08				4			2022-05	9	0	0
2020-03	2019-06	9	2	7	1,455,000	0	2022-05	26	0	0
							2021-03		0	0
2020-06	2020-01	5	1	6	2,956,181	1	2021-04	10	0	0
2019-10	2019-07	3	1	5	14,400,038	0	2021-09	23	0	0
							2021-05		0	0
									0	0
2020-09	2016-10	47	4	2	300,000	0	2021-12	15	0	0
2022-04	2021-12	4	1	4	12,890,000	0			0	0
2021-12	2021-01	11	1	1	2,470,000	0			0	0
2020-05	2020-05	0	2	2	0	0			0	0
							2022-09		0	0
2022-12	2022-04	8	9	10	14,193,179	0			0	0
2018-02	2017-12	2	1	2	2,200,000	0			0	0
2018-03	2016-10	17	1	4	792,000	0	2020-10	31	0	0
2019-08	2019-07	1	1	2	975,000	0	2022-10	38	0	0
2022-03	2021-01	14	1	1	1,750,000	0			0	0
2019-01	2016-12	25	3	3	4,400,000	1	2022-03	38	0	0
2022-09	2020-08	25	1	9	2,600,000	0			0	0
2020-09	2019-02	19	7	12	6,000,000	0			0	0
2019-12	2018-11	13	1	2	4,005,063	0	2021-10	22	0	0
2022-09	2021-04	17	6	9	19,900,000	0			0	0
2021-06				1					0	0
2019-05	2017-12	17	4	1	21,263,056	1	2022-06	37	0	0
2022-09	2021-11	10	1	2	20,700,000	0			0	0
2022-10	2020-02	32	3	4	5,000,000	0			0	0
2019-04	2017-11	17	1	5	10,000,000	0			0	0
2021-06	2020-05	13	1	10	1,346,746	0			0	0

The dashboard below summarizes how we analyzed the various funding windows:

Type of investment	Success
A. Not enough time	-1
B. Didn't raise Series B	0
C. Raised non-top-tier Series B	0
D. Raised top-tier Series B	1

Success	# of data points
-1	601
0	431
1	59

DATA BELOW IS INPUT WE CAN USE TO CHECK RESULT DEPENDING ON SCENARIO

	Metric
Series A How Long Ago	**36**
Success months	**36**
% Eligible	45%
% Successful Cos.	31%
% Successful Cos. On Timeline	21%

	# Months						
	12	18	24	30	36	48	60
% of companies Eligible	80%	71%	57%	52%	45%	30%	19%
Out of eligibles, % that is successful	3%	6%	12%	16%	21%	40%	67%

Total Companies = 1,033

Definition of companies Eligible: Companies that have raised a Series A in less than X months ago from today (# of months in the column)

% that is successful: Companies that have raised Series B from a Top Tier investor in the # of months in the column after the Series A

The image below illustrates how we consolidated the company and funding datasets using “Fuzzy Lookup” in Excel based on the provided single key (a small sample size was discarded in this process as some companies were impossible to match).

	A	B	C	D
1	Company - Funding	Company - Company	Similari	Company_adjusted
790	Zive	Zive	1.00	Zive
791	Zoe Financial	Zoe Financial	1.00	Zoe Financial
792	Zoop	Zoop	1.00	Zoop
793	Key Data Dashboardâ„¢	Key Data Dashboard	0.99	Key Data Dashboard
794	Light Field Lab	Light Field Lab, Inc.	0.98	Light Field Lab, Inc.
795	Maxwell Financial Labs	Maxwell Financial Labs, I	0.98	Maxwell Financial Labs, Inc.
796	Building Intelligence	Building Intelligence Inc.	0.98	Building Intelligence Inc.
797	Energetic Insurance	Energetic Insurance, Inc.	0.98	Energetic Insurance, Inc.
798	Julia Computing	Julia Computing, Inc.	0.98	Julia Computing, Inc.
799	Magnetic Insight	Magnetic Insight Inc	0.98	Magnetic Insight Inc
800	Neat Capital	Neat Capital Inc.	0.98	Neat Capital Inc.
801	Paperless Parts, Inc.	Paperless Parts	0.98	Paperless Parts
802	Virtual Peaker	Virtual Peaker, Inc	0.98	Virtual Peaker, Inc
803	Wahed Invest	Wahed Invest Inc.	0.98	Wahed Invest Inc.
804	Catch&Release	Catch&Release, Inc.	0.97	Catch&Release, Inc.
805	Bear Robotics	Bear Robotics, Inc.	0.97	Bear Robotics, Inc.
806	GRO Biosciences	GRO Biosciences, Inc.	0.97	GRO Biosciences, Inc.
807	Picus Security	Picus Security Inc.	0.97	Picus Security Inc.
808	Rafay Systems	Rafay Systems, Inc.	0.97	Rafay Systems, Inc.
809	Realtime Robotics	Realtime Robotics, Inc.	0.97	Realtime Robotics, Inc.
810	Soft Robotics	Soft Robotics Inc.	0.97	Soft Robotics Inc.
811	Symbio Robotics	Symbio Robotics, Inc.	0.97	Symbio Robotics, Inc.
812	Sidekick Health	SidekickHealth	0.97	SidekickHealth
813	CareAcademy	Care Academy	0.97	Care Academy
814	Mio	M.io	0.97	M.io
815	NavigatorCRE	Navigator CRE	0.97	Navigator CRE
816	ReturnLogic	Return Logic	0.97	Return Logic
817	Terminal49	Terminal 49	0.97	Terminal 49
818	ThreatX	Threat X	0.97	Threat X
819	UnitedMasters	United Masters	0.97	United Masters
820	Wildtype	Wild Type	0.97	Wild Type
821	Click Therapeutics	Click Therapeutics, Inc.	0.97	Click Therapeutics, Inc.
822	Cover Technologies	Cover Technologies, Inc.	0.97	Cover Technologies, Inc.
823	GraphWear Technologies	GraphWear Technologies	0.97	GraphWear Technologies Inc.
824	Medicom Technologies	Medicom Technologies, I	0.97	Medicom Technologies, Inc.
825	Nikola Labs Inc.	Nikola Labs	0.97	Nikola Labs
826	Phiar Technologies	Phiar Technologies, Inc	0.97	Phiar Technologies, Inc

Snapshots of the raw data set:

Funding data:

uuid	name	type	permalink	cb_url	rank	created_at	updated_at	country_code	state_code	region	city	investment_type	investor_count	announced_on	raised_amount_usd	raised_amount	raised_amount_currency_code	post_money_valuation_usd	post_money_valuation	post_money_valuation_currency_code
7060f160	Series D -	funding_r	solicore-se	https://ww	387666	9/26/2008 19:17	2/12/2018 22:25	USA	WA	Washingt	Redmond	series_d	6	7/27/2009	1.2E+07	1.2E+07	USD			
10fa177c-	Angel Rou	funding_r	stocktwits	https://ww	380145	12/17/2008 11:03	2/12/2018 23:57	USA	NY	New York	New York	angel	7	9/23/2008						
450eff0e-	Seed Rour	funding_r	stocktwits	https://ww	186687	12/18/2008 21:41	2/12/2018 23:27	USA	NY	New York	New York	seed	6	12/17/2008	800000	800000	USD			
f6512511	Seed Rour	funding_r	arzeda-se	https://ww	240285	4/14/2009 17:14	2/12/2018 23:22	USA	WA	Washingt	Seattle	seed	1	2/27/2009						
ee6cea19-	Series B -	funding_r	conductor	https://ww	274699	4/29/2009 20:15	2/12/2018 23:45	USA	NY	New York	New York	series_b	2	4/29/2009	1E+07	1E+07	USD			
7b4af7e6-	Seed Rour	funding_r	stocktwits	https://ww	396767	5/18/2009 17:29	2/12/2018 23:41	USA	NY	New York	New York	seed	1	5/18/2009	800000	800000	USD			
7e24833b	Series A -	funding_r	conductor	https://ww	61024	7/23/2009 13:23	2/12/2018 23:44	USA	NY	New York	New York	series_a	2	12/26/2006	2600000	2600000	USD			
f641332e-	Series A -	funding_r	recurve-se	https://ww	157653	7/29/2009 12:41	2/12/2018 23:32	USA	CA	California	San Franci	series_a	2	9/23/2008	6000000	6000000	USD			
2ba21280	Debt Finar	funding_r	solicore-d	https://ww	138482	9/9/2009 11:33	2/12/2018 23:23	USA	WA	Washingt	Redmond	debt_finar	1	6/3/2008	5000000	5000000	USD			
eeb7fa33-	Venture R	funding_r	mocapay-	https://ww	330244	9/28/2009 17:41	2/12/2018 23:00	USA	CO	Colorado	Denver	series_unl	2	9/16/2009	3000000	3000000	USD			
49164784	Series A -	funding_r	stocktwits	https://ww	231211	12/2/2009 11:48	2/12/2018 23:40	USA	NY	New York	New York	series_a	2	12/1/2009	3000000	3000000	USD			
de131ca4-	Series D -	funding_r	solicore-se	https://ww	280449	12/28/2009 15:45	2/12/2018 22:36	USA	WA	Washingt	Redmond	series_d	4	12/28/2009	1100000	1100000	USD			
93aab93f-	Seed Rour	funding_r	ucan-seed	https://ww	383999	12/30/2009 6:15	7/10/2020 7:51	USA	CT	Connectic	Woodbrid	seed		12/28/2009	1859316	1859316	USD			
7f7fe758-	Seed Rour	funding_r	veritonic-i	https://ww	26790	10/7/2020 16:06	1/21/2022 18:34	USA	NY	New York	New York	seed	4	10/7/2020	3200000	3200000	USD			
168a8081	Series A -	funding_r	mocapay-	https://ww	301363	3/2/2010 12:31	2/12/2018 23:39	USA	CO	Colorado	Denver	series_a	1	7/12/2007	3250000	3250000	USD			
cfe722b6-	Series B -	funding_r	telltale-ga	https://ww	368083	3/18/2010 13:18	2/12/2018 23:27	USA	CA	California	San Rafael	series_b	2	6/12/2007	6000000	6000000	USD			
f6886894-	Seed Rour	funding_r	telltale-ga	https://ww	167020	6/11/2010 13:12	2/12/2018 23:31	USA	CA	California	San Rafael	seed	1	2/28/2006	1400000	1400000	USD			
035faa9d-	Venture R	funding_r	recurve-se	https://ww	41000	6/18/2010 3:46	2/13/2018 0:47	USA	CA	California	San Franci	series_unl	1	6/17/2010	8000000	8000000	USD			
f2e05ab4-	Series C -	funding_r	nanosys-s	https://ww	94161	10/1/2010 8:52	1/13/2021 6:10	USA	CA	California	Milpitas	series_c	16	11/9/2005	4.2E+07	4.2E+07	USD			
3982c1a0	Venture R	funding_r	stocktwits	https://ww	312386	10/19/2010 13:05	2/12/2018 23:43	USA	NY	New York	New York	series_unl	2	10/19/2010	4000000	4000000	USD			
46c06ad9	Seed Rour	funding_r	wespire-s	https://ww	143785	11/5/2010 3:05	2/12/2018 23:55	USA	MA	Massachu	Boston	seed	1	11/4/2010	750000	750000	USD			
de6a158b	Venture R	funding_r	videon-ce	https://ww	176867	11/17/2010 15:33	2/12/2018 23:49	USA	PA	Pennsylva	State Colle	series_unl	1	11/16/2010						
c5357ccf-	Series B -	funding_r	panzura-s	https://ww	393713	12/8/2010 2:52	2/12/2018 23:40	USA	CA	California	San Jose	series_b	3	12/7/2010	1.2E+07	1.2E+07	USD			
3f3504a4-	Angel Rou	funding_r	materna-r	https://ww	379347	2/17/2011 0:09	2/12/2018 23:56	USA	CA	California	Mountain	angel	2	12/1/2010	1000000	1000000	USD			
536cf65e-	Seed Rour	funding_r	kevel-seec	https://ww	119384	3/17/2011 19:42	2/12/2018 23:05	USA	NC	North Car	Durham	seed	6	7/12/2011	650000	650000	USD			
989f81ab-	Pre Seed F	funding_r	minomons	https://ww	342688	3/18/2011 7:13	11/25/2020 11:24	CAN	QC	Quebec	MontrÃ©.	pre_seed	1	1/1/2011	18000	18000	USD			
7dcc838a-	Debt Finar	funding_r	minomons	https://ww	117034	3/19/2011 21:46	11/25/2020 11:44	CAN	QC	Quebec	MontrÃ©.	debt_finar	1	2/1/2011	150000	150000	USD			
1f2b82af-	Pre Seed F	funding_r	nestio-pre	https://ww	74873	4/14/2011 2:39	12/28/2020 17:39	USA	NY	New York	New York	pre_seed	2	1/4/2011	120000	120000	USD			
1cfc2222-	Angel Rou	funding_r	nestio-ang	https://ww	234277	5/2/2011 23:15	5/28/2021 10:18	USA	NY	New York	New York	angel	1	2/1/2010	55000	55000	USD			
71fb7885	Series B -	funding_r	smartcar-	https://ww	1863	1/21/2022 2:07	1/21/2022 2:07	USA	CA	California	Mountain	series_b	3	1/19/2022	2.4E+07	2.4E+07	USD			
634a8f8b-	Series C -	funding_r	telltale-ga	https://ww	285862	6/8/2011 2:35	2/12/2018 23:46	USA	CA	California	San Rafael	series_c	2	6/7/2011	7000000	7000000	USD			
f9b76f9c-	Angel Rou	funding_r	nestio-ang	https://ww	249258	7/6/2011 0:34	2/12/2018 23:55	USA	NY	New York	New York	angel	6	7/5/2011	750000	750000	USD			
441e1711	Series C -	funding_r	high-stree	https://ww	390486	8/11/2011 2:19	2/12/2018 23:37	USA	MA	Massachu	Boston	series_c	3	8/10/2011	6300000	6300000	USD			
9c6cc019-	Grant - Na	funding_r	nanosys-g	https://ww	222456	8/13/2011 3:52	2/12/2018 23:37	USA	CA	California	Milpitas	grant	1	8/12/2011	1.1E+07	1.1E+07	USD			

Company data:

uuid	permalink	company_name	primary_role	homepage_url	twitter_url	facebook_url	linkedin_url	short_description	domain	country_code	category_list	founded_on	funding_rounds	aliases	city	funding_total_usd	cb_url	employee_count	status	category_group_list	address	roles
6f6ee6f4-	0x-project	0x	company	https://0x	https://twitter.com/	https://ww	The proto	0xproject.	USA			#######	3		San Franci	2.4E+07	https://ww	10-Jan	operating			company
76f92407-	1eq	Babyscript	company	http://ww	https://tw	https://ww	An Enhanc	getbabysc	USA		Child Care	3/8/2013	5	1eq Inc.	Washingt	9795112	https://ww	Nov-50	operating	Apps,Heal	1101 30th	company
06585e67-	2u-laundr	2ULaundr	company	https://ww	https://tw	https://ww	2ULaundr	2ulaundry	USA		E-Comme	#######	2		Charlotte	2900000	https://ww	Nov-50	operating	Clothing a	1700 Mec	company
515185d7	3am-inno	3AM Inno	company	http://ww	https://tw	https://ww	3AM has c	3aminnov	USA			7/1/2015	1		Buffalo	537500	https://ww	10-Jan	operating		77 Goode	company
f492ce23-	3dlook	3DLOOK	company	https://ww	https://tw	https://ww	3DLOOK d	3dlook.me	USA		3D Techno	5/1/2016	5		San Francisco		https://ww	Nov-50	operating	Artificial Intelligence		company
e99d7593	3t-bioscie	3T Bioscie	company	http://ww	https://twitter.com/	https://ww	3T Bioscie	3tbioscien	USA			1/1/2017	2		Menlo Par	1.2E+07	https://ww	10-Jan	operating			company,
0fa81e32-	410-medic	410 Medic	company	http://41(	https://twitter.com/	https://ww	410 Medic	410medic	USA		Medical D	1/1/2013	4		Durham	7574993	https://ww	10-Jan	operating	Health Ca	201 West	company
8e5f9a48-	5thcolum	5thColum	company	http://5th	https://ww	https://ww	5thColum	5thcolum	USA		Big Data,D	4/1/2012	0		Chicago		https://ww	Nov-50	operating	Artificial I	640 N. La	company
ff6ad078-	75-fahren	75FÂ®	company	http://75f	https://tw	https://ww	Predictive	75f.io	USA		Energy Eff	1/1/2012	4		Burnsville	2855000	https://ww	Nov-50	operating	Energy,Int	221 River	company
a39f6a72-	82-labs-in	82 Labs In	company	https://ww	https://tw	https://ww	82 Labs In	82labs.co	USA		Consumer	8/1/2017	3	Morning R	Los Angel	7807925	https://ww	10-Jan	operating	Consumer	3530 Wils	company
6a6a591f-	abgenomi	AbGenom	company	http://abgenomics.com			AbGenom	abgenomi	USA		Biotechno	6/1/2000	3	ABGENON	Los Altos	4E+07	https://ww	Nov-50	operating	Biotechno	4984 El Ca	company
f1c60fa6-	abodo	ABODO	company	https://ww	http://twi	https://ww	http://ww	ABODO is	abodo.co	USA	Local Adv	1/3/2011	5	MoveIn	Madison	7318000	https://ww	Nov-50	operating	Advertisin	316 West	company
94d0d181	abot-labs-	Abot Labs	company	https://www.abotlabs.com/				AI-powere	abotlabs.c	USA	Artificial I	#######	0		Los Angeles		https://ww	Nov-50	operating	Artificial Intelligence		company
1ac2f72c-	accelbyte-	AccelByte	company	https://ac	https://tw	https://ww	https://ww	Online gar	accelbyte.	USA	Developer	#######	0		Redmond		https://ww	unknown	operating	Gaming,Sc	16110 NE	company
ae331822	accern	Accern	company	https://ac	http://twi	http://ww	https://ww	Accern ma	accern.co	USA	Analytics,A	8/4/2014	3		New York	3535000	https://ww	Nov-50	operating	Artificial I	110 Wall S	company
a8b6100a	access-va	Access Va	company	http://www.accessvascularinc.		https://ww	Access Va	accessvas	USA		Medical D	1/1/2015	2		Bedford	3774000	https://ww	10-Jan	operating	Health Ca	175 Middl	company
cd0a14ae-	aclaimant	Aclaimant	company	https://ac	https://twitter.com/	https://ww	Aclaimant	aclaimant	USA		Internet	1/1/2013	1		Chicago	2956181	https://ww	Nov-50	operating	Internet S	20 WEST K	company
b35b78b6-	adaptive-p	Adaptive F	company				Adaptive	Phage Ther	USA		Pharmace	1/1/2017	1		Bethesda	765000	https://ww	unknown	operating	Health Ca	7211 EXP	company
b890e6f1-	adar-it	Nerdio	company	http://ww	https://tw	https://ww	https://ww	Nerdio is a	getnerdio.	USA	Cloud Con	1/1/1998	1		Skokie	2400000	https://ww	101-250	operating	Internet S	8001 Linc	company
58b04e9f-	adlumin-ir	Adlumin, I	company	https://ww	https://tw	https://www.facebo	Adlumin d	adlumin.c	USA		Enterprise	9/1/2016	1		Arlington	300000	https://ww	10-Jan	operating	Hardware	1109 BEVI	company
fbdb4430-	adranos-e	Adranos E	company	http://adranosenergetics.com/		https://ww	Adranos E	adranoser	USA		Aerospace		1		Lafayette	620000	https://ww	10-Jan	operating	Science ar	P.O. BOX	company
93fa5adb-	adtrib	MaestroQ	company	https://ww	https://twitter.com/	https://ww	Quality As	maestroq	USA		Call Cente	1/1/2013	2	AdTrib,Ma	New York	1750000	https://ww	10-Jan	operating	Administrative Servic		company
2b7ebcf0-	advocado	Advocado	company	https://ww	https://tw	https://ww	https://ww	AI, SaaS, M	advocado	USA	Advertisin	1/1/2018	0	Advocado	Saint Louis		https://ww	10-Jan	operating	Advertisin	50 Maryla	company
1db5c88a-	adzerk	Adzerk	company	https://ad	http://twi	http://ww	http://ww	Adzerk hel	adzerk.co	USA	Ad Server,	1/1/2009	1		Durham	650000	https://ww	Nov-50	operating	Advertisin	501 Wash	company
bf64268b-	aella-data	Aella Data	company	https://ww	https://twitter.com/	https://ww	Aella Data	aelladata.	USA		Artificial I	1/1/2015	1		Santa Clara		https://ww	Nov-50	operating	Artificial I	Building 2	company
8d09a443	aerones	Aerones	company	https://ae	https://tw	https://www.facebo	Industrial	aerones.c	USA		Drone Ma	1/1/2015	2		San Jose	541610	https://ww	10-Jan	operating	Consumer Electronic		company
e125554a-	agentdesk	Agentdesk	company	http://ww	http://twi	http://ww	https://ww	Agentdesk	agentdesk	USA	CRM,Mob	#######	4		San Franci	6504000	https://ww	Nov-50	operating	Community and Lifes		company
05becba9-	agent-iq	Agent IQ	company	http://age	https://tw	https://ww	https://ww	Agent IQ i	agentiq.co	USA	Artificial I	#######	3	Agent IQ,A	San Franci	8500000	https://ww	10-Jan	operating	Artificial I	44 Teham	company
d3280aba-	agility-rob	Agility Rob	company	http://ww	https://tw	https://ww	https://ww	Agility Rob	agilityrobo	USA	Industrial	1/1/2015	3		Albany	8000000	https://ww	10-Jan	operating	Hardware	421 Wate	company
a08d7799	agora-9	Agora	company	http://ww	https://tw	https://www.facebo	Efficient S	agoratix.c	USA		Analytics,A	1/1/2017	1		San Franci	680000	https://ww	10-Jan	operating	Commerce and Shop		company
7a61de5b	agrospher	AgroSpher	company	http://www.agrospheres.com/				AgroSpher	agrospher	USA		#######	7		Charlottes	750000	https://ww	10-Jan	operating		1180 Semi	company
88527c94-	agvend	AgVend	company	https://ww	https://tw	https://ww	AgVend is	agvend.co	USA		Agriculture,Compute		1		Minneapo	1750000	https://ww	10-Jan	operating	Agriculture and Farm		company
4fa325c8-	aifi-c8f8	AiFi	company	http://aifi.io		https://ww	https://ww	AiFi is a co	aifi.io	USA	Artificial I	1/1/2016	1		Santa Clar	4000000	https://ww	Nov-50	operating	Artificial I	Santa Clar	company
a6cb2c40-	aikon	Aikon	company	https://ail	https://tw	https://ww	https://ww	A softwar	aikon.com	USA	Computer	1/1/2017	1		San Mate	2000000	https://ww	10-Jan	operating	Consumer Electronic		company
03ef8031-	aingel	Aingel Cor	company	http://ain	https://twitter.com/AingelAI			Aingel is a	aingel.ai	USA	Artificial I	#######	2		San Jose	800000	https://ww	Nov-50	operating	Artificial I	75 E Santa	company
218451c9-	air	Air	company	http://airc	https://tw	https://ww	https://ww	Air offers	aircamera	USA	Cloud Mar	1/3/2017	1		Brooklyn		https://ww	10-Jan	operating	Informatic	325 Rutle	company
6bc21b9c-	aisense-in	AISense In	company	http://ww	https://twitter.com/	https://ww	AISense in	aisense.co	USA		Artificial I	2/1/2016	2		Los Altos	1.3E+07	https://ww	Nov-50	operating	Artificial Intelligence		company
7754b10e-	akademu-	Akademu	company	http://aka	https://tw	https://ww	https://ww	Akademu	akademu.	USA	Biotechno	1/1/2014	5		Ann Arbor	3105063	https://ww	10-Jan	operating	Biotechno	401 W. M	company
5bbdc03e-	alcatraz	Alcatraz A	company	https://ww	https://tw	https://ww	https://ww	Alcatraz A	alcatraz.a	USA	3D Techno	4/1/2016	2	Rainbe Inc	Palo Alto		https://ww	10-Jan	operating	Data and /	516 High S	company
18c2d160-	aliaswire	Aliaswire	company	https://ww	https://tw	https://ww	https://ww	A leading	aliaswire.c	USA	Financial S	1/1/2005	0		Burlington		https://ww	Nov-50	operating	Financial S	152 Middl	company